7.1.1 Querying unstructured data

Unstructured data is information that doesn't neatly fit into typical tables or databases. It's usually filled with text, but can also include things like dates, numbers, or facts. Examples of unstructured data are emails, documents, photos, social media posts, and more.

Suppose our search engine contains this quote from Albert Einstein:

```
"quote": "Imagination is more important than knowledge"
```

Users get positive results with queries consisting of either an individual word or a combination of words ("imagination", "knowledge", etc.). Table 7.1 shows a set of search keywords users can search with in this case and the expected results.

Table 7.1 Possible search queries and expected results

Search keywords	Results	Notes
"imagination", "knowledge"	Yes	Individual keywords are exact matches, so positive results are returned.
"imagination knowledge", "knowledge important"	Yes	Combined keywords also match the document and so return results.

However, searching for other criteria may yield no results. For example, as shown in table 7.2, if the user searches for "passion", "importance", "passionate wisdom", "curious cognizance", or similar terms, the engine fails to return matching results with the default settings. Words such as *passion* and *cognizance* are synonyms in the quote. (Similarly, the default settings miss abbreviations.) We can tweak the analyzers to honor synonyms, stemming, spelling mistakes, and more, as we see later in the chapter.

Table 7.2 Possible search queries with no results

Search keywords	Results	Notes
"imagine", "passion", "curious", "importance", "cognizance", "wisdom", "passionate wisdom", "extra importance"	No	Synonyms and alternate names do not yield positive results.
"impartant", "knowlege", "imaginaton"	No	Spelling mistakes may result in poor or no matches.
Imp, KNWL, IMGN	No	Abbreviations do not return positive results.

We should expect users to query our engine with numerous combinations: synonyms, abbreviations, acronyms, emojis, lingo, and so on. A search engine that helps fetch relevant answers for a variety of search criteria always wins.